Gmail Hariharan D <embeddedproductdev@gmail.com>

Documents for Universal Indoor Ductable Unit Controller (2 unit Type)

6 messages

Hariharan D <embeddedproductdev@gmail.com> Thu, Aug 13, 2026 at 2:54 PM
To: "sumanelectronicsthane@gmail.com" <sumanelectronicsthane@gmail.com>
Cc: srini.pcbdesigner@gmail.com, Matheen Matheen <matheen@unimationrobotics.com>

Hi,

We just talked over the phone. Kindly share the documents related to Universal Indoor Ductable Unit Controller (2 unit type) to assist with our development.

Regards,
Hariharan D

Suman Electronics <sumanelectronicsthane@gmail.com> Thu, Aug 13, 2026 at 3:45 PM
To: Hariharan D <embeddedproductdev@gmail.com>
Cc: srini.pcbdesigner@gmail.com, Matheen Matheen <matheen@unimationrobotics.com>

Hi,

Please find attached documents for reference.

Regards,
Neha Patil
Partner - Suman Electronics
[Quoted text hidden]

3 attachments

DIP SWITCH SETTINGS.pdf
195K

MODBUS DATA FORMAT VER 01.01.pdf
368K

COG_128_64_Handset_User_Manual.-2.pdf
808K

Hariharan D <embeddedproductdev@gmail.com> Thu, Aug 13, 2026 at 4:34 PM
To: Suman Electronics <sumanelectronicsthane@gmail.com>
Cc: srini.pcbdesigner@gmail.com, Matheen Matheen <matheen@unimationrobotics.com>

Hi Neha,

Received the documents. Thank you for sharing them.

Regards,
Hariharan D
[Quoted text hidden]

Srini Vasan <srini.pcbdesigner@gmail.com> Wed, Aug 26, 2026 at 8:46 PM

To: Hariharan D <embeddedproductdev@gmail.com>
Cc: Suman Electronics <sumanelectronicsthane@gmail.com>, Matheen Matheen <matheen@unimationrobotics.com>

Hi Neha,

Can you please share the wiring diagram for the Universal Indoor Ductable Unit Controller (2 unit type)?
Where do I need to connect the Power supply, communication lines and other lines?

Regards
Srini
[Quoted text hidden]

Hariharan D <embeddedproductdev@gmail.com> Thu, Aug 27, 2026 at 3:42 PM
To: Srini Vasan <srini.pcbdesigner@gmail.com>
Cc: Suman Electronics <sumanelectronicsthane@gmail.com>, Matheen Matheen <matheen@unimationrobotics.com>

Subject: Technical Queries — Modbus Data Format Document (Version 01/01) | AC Controller Integration

Dear Suman Electronics Team,

Greetings. We are currently integrating your AC Controller into our IoT-based Building Management System using the Modbus RTU protocol over RS485. As part of our firmware development, we have reviewed your Modbus Data Format document (Version 01/01, dated 13/06/2019) in detail.

While the document provides a good foundation, we have identified several gaps and apparent discrepancies that are blocking our firmware implementation. We would be grateful if you could provide clarifications on the following points at your earliest convenience.

CRITICAL — Blocking firmware implementation

Q1. Registers 11 & 22 — Temperature Value Format (RAT & SAT)
Registers 11 (Return Air Temperature) and 22 (Supply Air Temperature) have no documented unit, scaling factor, or valid range. Could you please confirm:
a) Is the value a raw integer in °C (e.g. 27 = 27°C), or scaled (e.g. 270 = 27.0°C × 0.1)?
b) What is the valid range of the register value?
c) What value is returned when the sensor is disconnected or in a fault condition?

Q2. Registers 25 & 26 — Compressor Current Value Format
No unit, scaling factor, or valid range is documented for Compressor 1 and Compressor 2 Current registers. Could you please confirm:
a) Is the value in Amps directly (e.g. 15 = 15A), or is it scaled (e.g. 150 = 15.0A × 0.1)?
b) What is the maximum expected current value in the register?

Q3. Register 16 Note — Apparent Typo Regarding Fan Registers
The document states: "Register number 22 is setting register while register number 12 is actual status of blower fan." However, Register 22 is Supply Air Temperature and Register 12 is Compressor 1 Run Hours — neither is related to the fan. Could you please clarify:
a) Which register is the fan speed setting register?
b) Which register reflects the actual fan status?

Q4. Register 3 (RESET_COMMAND) — Reset Sequence
After writing 1 to Register 3 to reset a fault, we need to understand the expected sequence:
a) Should the PC write 0 again after the reset, or does the controller auto-clear Register 3?
b) How long should the PC wait before re-reading the fault registers to confirm the reset was successful?
c) What is the behaviour if Register 10 (FAULT_RESET_ACTIVE) is 0 when Register 3 is written as 1?

Q5. Registers 20 & 21 — D0 Bit Values Appear Identical (Possible Typo)
In Register 20 (Compressor 1 Fault), both D0=0 and D0=1 are documented as "ODU 1 SPPR CLEARED". We believe D0=1 should read "ODU 1 SPPR FAULT EXIST". The same issue appears in Register 21 for ODU 2. Could you please confirm the correct bit descriptions for D0 in both registers?

IMPORTANT — Required before field testing

Q6. Registers 12, 13, 14 — Run Hours Format and Persistence
a) Are these raw integer hours (16-bit, max 65535 hours) or is there a scaling factor?
b) What happens when the counter rolls over?
c) Are these values stored in EEPROM (survive power cycle) or reset on power-on?

Q7. Register 16 (BLOWER_FAN_STATUS) — D0 Behaviour on IDU Machines
D0 is documented for Package machines (fan on/off). For IDU machines, does D0 also indicate fan on/off alongside D1/D2 (speed), or is D0 always 0 on IDU machines?

Q8. Register 15 (COMPRESSOR_STATUS) — 3-Compressor Support
D2 is documented as Compressor 3 ON/OFF. However, the handset manual only shows 2 compressors. Is a 3-compressor configuration supported, and if so, how is it enabled?

Q9. Registers 23 & 24 — Humidity Label Error and Value Format
a) Both Register 23 and Register 24 appear to be labelled "HUMIDITY 1" in the register table. We believe Register 24 should be "HUMIDITY 2". Could you please confirm?
b) Is the humidity value a raw integer percentage (e.g. 52 = 52%) or scaled?

Q10. Register 2/9 (OPERATING_MODE) — Undefined Bits When Writing
D0, D1, D2, and D5 are defined. D3, D4, and D6–D15 are not documented. When writing Register 2, should all undefined bits be written as 0? Will non-zero values on undefined bits cause any unintended behaviour?

MINOR — Clarification and future-proofing

Q11. Register 6 — Read-Only vs Read/Write Discrepancy
Register 6 is listed as READ ONLY in the register table, but the bit details section implies it can be written by the PC to change the number of compressors. Could you please confirm whether Register 6 is read-only or read/write?

Q12. Register 7 (MODEL_TYPE) — D3 to D15
Only D0, D1, and D2 are defined. Are D3–D15 reserved and always 0?

Q13. Register 17 (ALARM_STATUS) — D1 to D15
Only D0 (alarm present/cleared) is defined. Do D1–D15 encode specific alarm categories, or are they reserved?

Q14. General — Register Values During Power-On Initialisation
What values do registers hold during the first 750ms after power-on, before the controller is ready to respond? Should the master wait for a specific ready condition or simply retry on timeout?

We would appreciate a response at your earliest convenience, as several of these points are on the critical path of our firmware development schedule.

Please feel free to reach out if you need any additional context about our integration or test setup.

Thank you for your support.

Warm regards,
Hariharan
Embedded Firmware Engineer
Project: Unimation IoT BMS — RS485 Modbus Integration
[Quoted text hidden]

Suman Electronics <sumanelectronicsthane@gmail.com> Wed, Sep 2, 2026 at 3:44 PM
To: Hariharan D <embeddedproductdev@gmail.com>
Cc: Srini Vasan <srini.pcbdesigner@gmail.com>, Matheen Matheen <matheen@unimationrobotics.com>

Hi,

Please find the response to the queries and also attached updated MODBUS document.

Q1. Registers 11 & 22 — Temperature Value Format (RAT & SAT)
Registers 11 (Return Air Temperature) and 22 (Supply Air Temperature) have no documented unit, scaling factor, or valid range. Could you please confirm:
a) Is the value a raw integer in °C (e.g. 27 = 27°C), or scaled (e.g. 270 = 27.0°C × 0.1)?
If the temperature value is 27.1, Register 11 and 22 will have 271.

b) What is the valid range of the register value?
Register 11 and 22 have temperature values read from sensors connected to the port. If no temperature sensor open is detected (from fault registers), values in these registers are to be considered as they are received.

c) What value is returned when the sensor is disconnected or in a fault condition?
Usually, it should be 0.

Q2. Registers 25 & 26 — Compressor Current Value Format
No unit, scaling factor, or valid range is documented for Compressor 1 and Compressor 2 Current registers. Could you please confirm:
a) Is the value in Amps directly (e.g. 15 = 15A), or is it scaled (e.g. 150 = 15.0A × 0.1)?
b) What is the maximum expected current value in the register?
Compressor current registers are applicable only if System with ODU is configured using its DIP switch. If configured as System Without ODU, these values should be ignored.

Q3. Register 16 Note — Apparent Typo Regarding Fan Registers
The document states: "Register number 22 is setting register while register number 12 is actual status of blower fan." However, Register 22 is Supply Air Temperature and Register 12 is Compressor 1 Run Hours — neither is related to the fan. Could you please clarify:
We will send an updated document to clarify this.

a) Which register is the fan speed setting register?
PC OPERATING MODE REGISTER (Register No. 2) is setting register.

b) Which register reflects the actual fan status?
BLOWER FAN STATUS REGISTER (Register No. 16) is an actual status register.

Q4. Register 3 (RESET_COMMAND) — Reset Sequence
After writing 1 to Register 3 to reset a fault, we need to understand the expected sequence:
a) Should the PC write 0 again after the reset, or does the controller auto-clear Register 3?
b) How long should the PC wait before re-reading the fault registers to confirm the reset was successful?
c) What is the behaviour if Register 10 (FAULT_RESET_ACTIVE) is 0 when Register 3 is written as 1?
For all above 3 questions:
FAULT_RESET_ACTIVE (Register 10) indicated when it is a valid condition to reset faults. If you write Register 3 as 1 when Register 10 is 0, it would be ignored by the controller. You should write 1 to Register 3 only if you get 1 in Register 10. Once this is received, the controller will reset all faults which could be cleared. So, you should write 1 to Register 3 only when you get 1 in register 10.

Q5. Registers 20 & 21 — D0 Bit Values Appear Identical (Possible Typo)
In Register 20 (Compressor 1 Fault), both D0=0 and D0=1 are documented as "ODU 1 SPPR CLEARED". We believe D0=1 should read "ODU 1 SPPR FAULT EXIST". The same issue appears in Register 21 for ODU 2. Could you please confirm the correct bit descriptions for D0 in both registers?
Yes, you're correct. We will correct this in the document and will resend it.

IMPORTANT — Required before field testing

Q6. Registers 12, 13, 14 — Run Hours Format and Persistence
a) Are these raw integer hours (16-bit, max 65535 hours) or is there a scaling factor?
It is a 16 bit integer value. It can go maximum up to 65535. There is no scaling factor.

b) What happens when the counter rolls over?

Once it rolls over, the controller will start counting from 0.

c) Are these values stored in EEPROM (survive power cycle) or reset on power-on?
Run Hours are saved in the controller's non volatile memory every 30 minutes. Controller will read these values on each Power ON.

Q7. Register 16 (BLOWER_FAN_STATUS) — D0 Behaviour on IDU Machines
D0 is documented for Package machines (fan on/off). For IDU machines, does D0 also indicate fan on/off alongside D1/D2 (speed), or is D0 always 0 on IDU machines?
D0 is applicable only in Package machines. It should be ignored in IDU machines. So, you should read D1 and D2 bits only.

Q8. Register 15 (COMPRESSOR_STATUS) — 3-Compressor Support
D2 is documented as Compressor 3 ON/OFF. However, the handset manual only shows 2 compressors. Is a 3-compressor configuration supported, and if so, how is it enabled?
3rd compressor is only applicable in Package machines. For IDU machines, this cannot be activated.

Q9. Registers 23 & 24 — Humidity Label Error and Value Format
a) Both Register 23 and Register 24 appear to be labelled "HUMIDITY 1" in the register table. We believe Register 24 should be "HUMIDITY 2". Could you please confirm?
Yes, you are right. We will send updated document for this.

b) Is the humidity value a raw integer percentage (e.g. 52 = 52%) or scaled?
This is the raw integer percentage. Please note that for reading Humidity, you will have to connect sensors supplied by us to ports. Also, these values are only for monitoring purposes.

Q10. Register 2/9 (OPERATING_MODE) — Undefined Bits When Writing
D0, D1, D2, and D5 are defined. D3, D4, and D6–D15 are not documented. When writing Register 2, should all undefined bits be written as 0? Will non-zero values on undefined bits cause any unintended behaviour?
Yes, all undefined bits should be 0. Writing these bits to 1 will cause the controller to ignore the command.

MINOR — Clarification and future-proofing

Q11. Register 6 — Read-Only vs Read/Write Discrepancy
Register 6 is listed as READ ONLY in the register table, but the bit details section implies it can be written by the PC to change the number of compressors. Could you please confirm whether Register 6 is read-only or read/write?
For this model, these cannot be written from a PC. So, this is a Read Only value for PC.

Q12. Register 7 (MODEL_TYPE) — D3 to D15
Only D0, D1, and D2 are defined. Are D3–D15 reserved and always 0?
Yes, D3 to D15 are always 0.

Q13. Register 17 (ALARM_STATUS) — D1 to D15
Only D0 (alarm present/cleared) is defined. Do D1–D15 encode specific alarm categories, or are they reserved?
D1 to D15 are always 0. This register is only for indication that the controller has sensed a fault.

Q14. General — Register Values During Power-On Initialisation
What values do registers hold during the first 750ms after power-on, before the controller is ready to respond? Should the master wait for a specific ready condition or simply retry on timeout?
Controller will communicate with the PC as soon as serial port initialisation is executed on power on. It should take less than a few milliseconds from power on. Rest all values would be updated as other peripherals are initialized. Generally, this also should not take more than about 500 milliseconds. So, there is no specific condition for which master should wait on power on. Master should follow its regular retry and timeout strategy.

Regards,
Sandeep S. Killedar
R & D - Suman Electronics
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MODBUS DATA FORMAT VER 01.02.docx
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